

EBX Quantitative Analysis

Development Research Report

Final reproducible development package

August 2026

Abstract

This report documents the EBX masked high-frequency-data analysis for Parts 1–5 of the challenge. The specified development universe contains 85 days, but the available and used development data contains 70 days: Days 1–64 and 80–85. Days 65–79 are explicit missing days. Days 86–108 are reserved holdout days and are not used for the development conclusions in this report.

The analysis finds non-normal, heavy-tailed returns; persistent or inconclusive intraday regimes under the declared tests; substantial feature redundancy; and modest predictive association that does not survive the documented transaction cost assumption in the fixed Part 5 baseline. Ridge is the strongest tested model by predictive association and after-cost development P&L. No claim of profitability, production readiness, causal feature identity, or holdout generalization is made.

1 Scope, data boundary, and reproducibility

The challenge specifies Days 1–85 for development. The repository freeze records:

Item	Value
Expected development days	85
Available development days	70
Missing development days	15 (Days 65–79)
Available day ranges	1–64 and 80–85
Reserved holdout	Days 86–108

All day-local transformations preserve day boundaries. Structural warm-up missingness is retained and distinguished from unexpected missingness; no silent imputation is used. The principal evidence is stored in the canonical CSV/JSON artifacts under `results/`, with the development freeze in `results/freeze/development_freeze.json`.

The holdout boundary is explicit: Days 86–108 are reserved holdout days and were not accessed, analyzed, or used for model selection, validation, or performance claims in the development experiments.

2 Part 1: data hygiene and descriptive statistics

The ingestion and validation pipeline checks filenames and day identity, schema, timestamps, duplicates, gaps, price validity, and structural missingness. The available data is processed as 70 independent daily files. Unexpected missingness is reported rather than silently repaired. The resulting development artifacts include the dataset inventory, quality summary, descriptive statistics, ACF diagnostics, and missingness tables.

The intraday volatility profile has a U-shaped pattern. RMS one-second volatility is approximately 6.1×10^{-5} in hour 0, falls to 3.8×10^{-5} around hour 3, and rises to 4.9×10^{-5} by hour 6. The low-lag return autocorrelation is generally small, consistent with weak microstructure dependence at the one-second horizon.

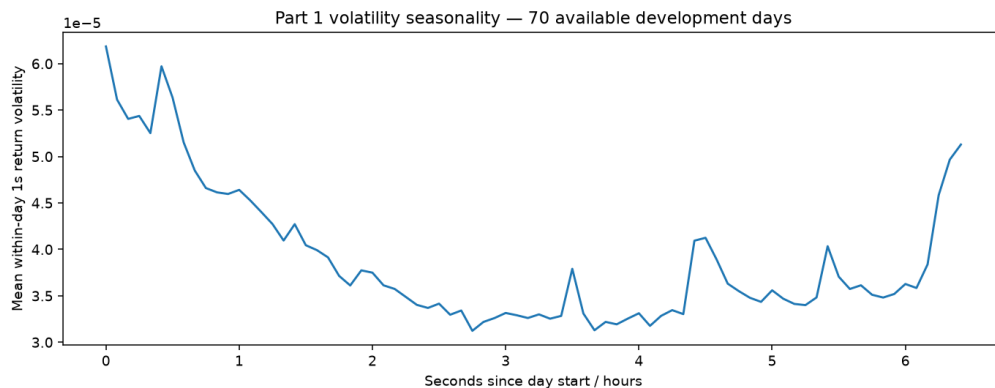


Figure 1: Available-development-day intraday volatility seasonality.

3 Part 2: distribution and tail analysis

The pooled one-minute and five-minute return distributions reject normality under the declared tests. Excess kurtosis is 16.2943 at one minute and 11.4267 at five minutes. Relative to a fitted Gaussian benchmark, the observed-to-expected ratios for observations beyond 3σ are 6.2054 and 5.9569, respectively. These results make a naive Gaussian risk model materially understate tail frequency.

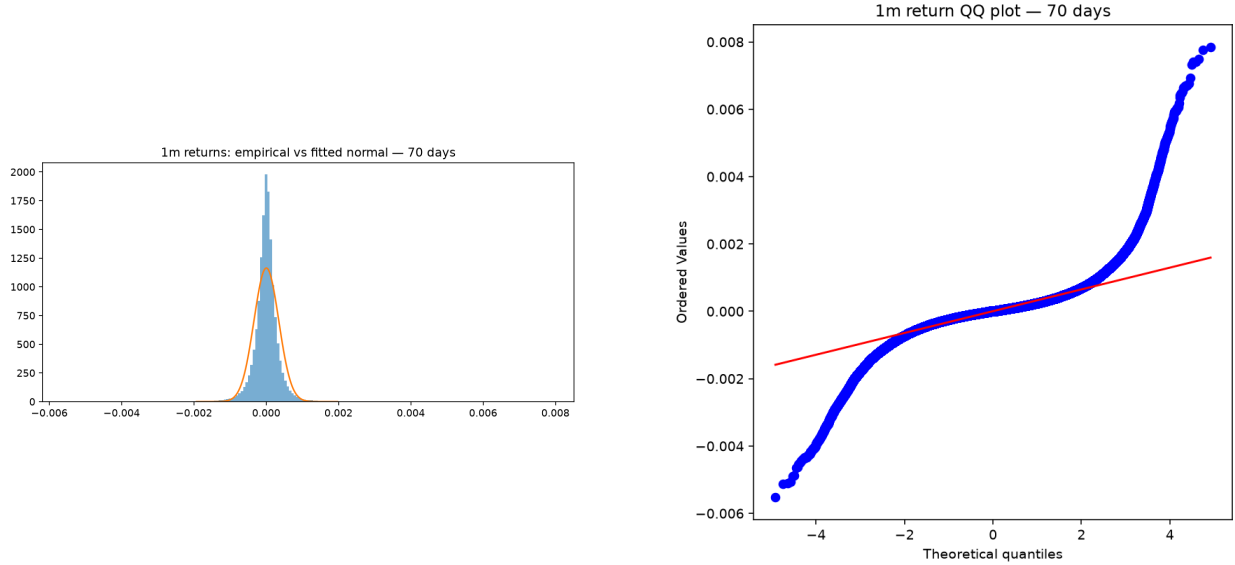


Figure 2: One-minute return distribution and normal QQ diagnostic.

The rare-event catalogue contains the most extreme one-minute moves available in the development scope. Volume coincidence is not asserted because the repository does not contain a validated raw-volume semantic source. The distribution, sigma-event, and tail-estimate artifacts are descriptive and do not constitute a production risk model.

4 Part 3: regime classification

The regime classifier combines variance-ratio, Hurst, autocorrelation, and ADF-based evidence under predeclared thresholds. Among the 70 classified development days, 61 are classified as momentum/persistent and 9 as random-walk/inconclusive; no available day is classified as mean-reverting. The 15 missing days remain explicit rows in the 85-row regime table and are not fabricated classifications.

The tests do not always agree. Day 84 is a clear example: it is retained in the primary model result but is classified as random-walk/inconclusive with low confidence under the frozen regime rule. Regime labels are descriptive state classifications, not evidence of a profitable strategy.

5 Part 4: feature forensics

The 691 masked features are organized into PB, VB, BB, PV, and V families. For fixed indicator index i , changing a suffix such as PB1_T1 to PB1_T2 is treated as a window variant within the same naming subfamily. This is a taxonomy statement, not source-code provenance.

The reverse-engineering pipeline evaluates hand-built price and return candidate formulas. The stored results support hypotheses, not recovery of the original feature generator. Redundancy analysis shows a highly dependent feature panel and a low-dimensional effective structure. The frozen predictive screen contains 543 feature-horizon rows. These are statistical associations; they

are not causal identities.

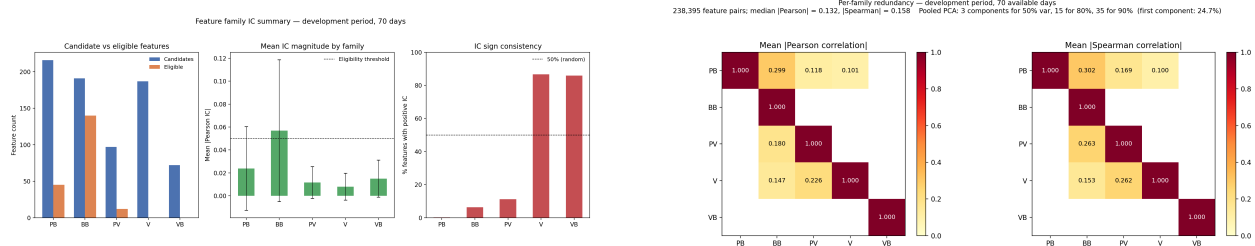


Figure 3: Feature-family predictive summary and redundancy evidence.

5.1 BB verification and provenance limitation

The taxonomy contains 83 features in the BB10–BB26 block: BB10–BB15 with temporal suffixes and BB16–BB26 as flat columns. The canonical repository artifacts do not contain an exact-match computation that establishes a single candidate formula reproducing all 83 columns from raw Price to machine precision. The existing candidate results instead report hypothesis-level matches using a finite candidate library. Therefore this report does not claim that the original BB generator has been recovered.

The comparison is not causal/source-generation evidence. It compares supplied masked values with hand-built formulas on available development data. The original feature-generation source is unavailable, so the provenance limitation remains unchanged. In particular, the 83-column family count must not be presented as a proof of causal construction or as confirmation of an exact indicator formula.

6 Machine-learning development comparison

The model-ready pipeline uses the exact within-day 300-second future return, training-only preprocessing, and chronological development splits. The frozen screen baseline uses 197 features. The training-only selection experiment selects 198 features from the 691-feature candidate universe, retaining the 197 frozen features plus one additional feature. The feature screen itself was predetermined by the existing Part 4 artifact for the frozen baseline; this is not claimed as a completely train-only feature-selection experiment.

Table 1: Primary W3 development model comparison.

Model	Pearson IC	Directional accuracy	Net P&L
Ridge	0.0707122893	0.5076512552	-0.2247185557
Elastic Net	0.0336523173	0.5082094429	-0.2431015522
LightGBM	0.0433187087	0.5213608887	-0.2445651284

Ridge is strongest by Pearson IC and net development P&L. Elastic Net and LightGBM do not improve the conclusion. All three comparisons are development experiments only; none establishes alpha, profitability, or holdout generalization.

7 Part 5: fixed baseline strategy and development backtest

The sole strategy is sign-based and deterministic: a positive prediction is long, a negative prediction is short, and zero is flat. Exposure is fixed at unit notional with one position at a time. Entry is the first same-day raw observation at or after the signal, and exit is the exact same-day timestamp 300 seconds after entry. The cost assumption is 5 basis points at entry plus 5 basis points at exit, with zero fee. This is an explicit parameterized assumption, not an empirically measured spread model.

Table 2: Development backtest results by chronological window.

Window	Trades	Gross P&L	Costs	Net P&L	Sharpe
W1	410	0.008677675	0.410000	-0.401322325	-186.162764
W2	285	0.014409138	0.285000	-0.270590862	-32.994427
W3	246	0.021281444	0.246000	-0.224718556	-106.280913
Pooled	941	0.044368257	0.941000	-0.896631743	-52.575858

Gross P&L is positive in each window, but the fixed transaction-cost assumption dominates it. The pooled result is descriptive because temporal windows reuse validation days across experiments. Day 84 remains in the primary W3 result. Excluding Day 84 is a post-hoc sensitivity only and reduces W3 Pearson IC from 0.070712 to 0.011037; it does not change the primary result or retrain the model.

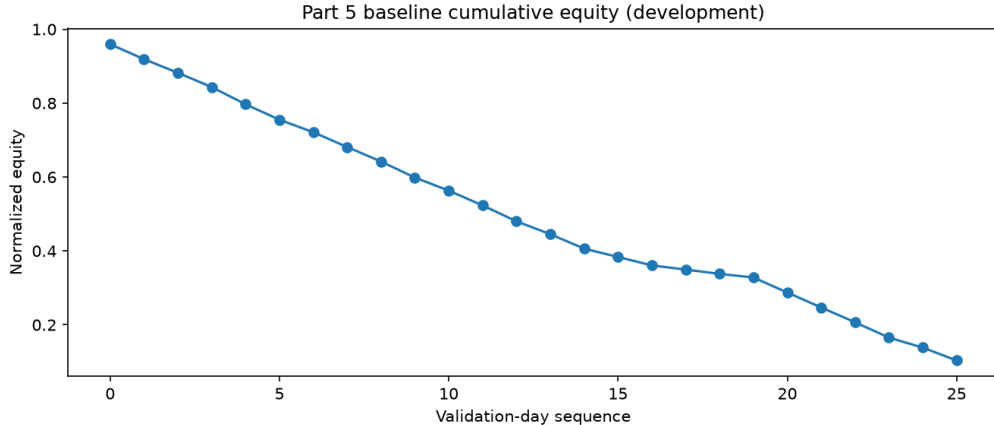


Figure 4: Fixed-rule development equity curve after documented costs.

8 Limitations and conclusion

The principal limitations are the 70-of-85-day development coverage, the explicit Days 65–79 gap, unavailable validated raw-volume semantics, masked feature provenance, negative after-cost development P&L, and the sensitivity of W3 predictive association to Day 84. The feature formulas remain hypotheses unless independently supported by source-generation evidence.

The final development conclusion is conservative: the repository provides reproducible statistical

and predictive-association evidence, but the fixed baseline does not demonstrate economic utility after costs. Ridge is the strongest tested model in the available development experiments. Days 86–108 are reserved holdout days and were not accessed, analyzed, or used for model selection, validation, or performance claims in the development experiments.